Effect of Preprocessing Choices, Model Complexity, and Regularisation Strategies on Generalisation Performance in Seizure Prediction Tasks

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Abstract

This paper presents a systematic experimental investigation into how preprocessing pipeline design, regularisation strategy, and class imbalance handling affect the generalisation performance of Logistic Regression models on epileptic seizure prediction tasks. Three real-world EEG datasets are used: the UCI Epileptic Seizure Recognition dataset (11,500 samples, 178 time-series features, 1:4 imbalance), the EEG-FREE Clinical Features dataset (4,500 samples, 33 clinical features, 2.4:1 imbalance), and the Statistical EEG Features dataset (200 samples, 11 statistical features, 1:1 balance). Two distinct preprocessing pipelines are compared: Pipeline A (StandardScaler → SelectKBest) and Pipeline B (MinMaxScaler → PCA). Regularisation methods L1 (Lasso), L2 (Ridge), and Elastic Net are evaluated across 27 experimental configurations (3 datasets × 3 regularisers × 3 imbalance strategies). Results demonstrate that preprocessing order significantly affects performance on high-dimensional and imbalanced datasets, L2 regularisation provides the most consistent generalisation, and class weighting offers a practical and effective imbalance correction. The study also demonstrates intentional overfitting and underfitting behaviour via validation and learning curves.

Index Terms—Seizure prediction, EEG classification, Logistic Regression, regularisation, preprocessing pipelines, class imbalance, SMOTE, Precision-Recall AUC.

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making it one of the most prevalent neurological disorders [1]. Automated seizure prediction from electroencephalogram (EEG) signals has been a key goal of clinical neuroscience and machine learning research. Accurate and timely detection of seizure onset can enable preventive interventions, improve patient quality of life, and reduce the burden on clinical staff.

However, the challenge of seizure prediction from EEG data extends well beyond model architecture. The choices made during preprocessing, including normalisation strategy, dimensionality reduction, and feature selection order, can substantially change the signal presented to the model. Similarly, regularisation strategy directly controls model complexity and its ability to generalise to unseen data. Class imbalance, a near-universal characteristic of seizure datasets, further complicates evaluation and model training.

This study focuses on Logistic Regression as the core classifier. Despite its simplicity, Logistic Regression is widely used in clinical ML due to its interpretability, probabilistic output, and strong performance when paired with appropriate preprocessing. The goal is not simply to achieve high accuracy, but to experimentally analyse and interpret how each design choice shapes model behaviour.

The four primary research questions investigated are: (1) Does preprocessing order affect model performance? (2) Which regularisation method generalises best across datasets? (3) Does Elastic Net consistently outperform L1 and L2? (4) How does class imbalance handling interact with regularisation?

II. DATASETS

Three publicly available epileptic seizure datasets were selected to provide complementary coverage of dataset size, imbalance level, and feature type.

A. DS1 — UCI Epileptic Seizure Recognition

The UCI Epileptic Seizure Recognition dataset contains 11,500 samples with 178 EEG time-series amplitude features (X1 to X178), each representing a one-second EEG segment. The original five-class label (y = 1: seizure; y = 2–5: non-seizure brain states) was binarised so that y = 1 represents seizure and y = 0 represents all other states. This produces a class imbalance ratio of 1:4 (2,300 seizure vs. 9,200 non-seizure samples). DS1 is the largest and most established benchmark in this study, with high-dimensional raw EEG features that make it suitable for testing feature selection and regularisation strategies.

B. DS2 — Clinical EEG Features (EEG-FREE)

The Clinical EEG Features dataset contains 4,500 samples with 33 mixed features including demographic information (age, gender), clinical symptoms, and seizure history. The target column was binarised: epilepsy of any type (Focal, Generalised, Structural, Unknown) versus No Epilepsy. This results in a 2.4:1 imbalance with 3,169 epilepsy samples and 1,331 non-epilepsy samples. Categorical features were encoded with Label Encoding prior to modelling. This dataset tests the model's behaviour on structured clinical data rather than raw EEG waveforms.

C. DS3 — Statistical EEG Features

The Statistical EEG Features dataset contains 200 samples with 11 statistical features extracted from EEG segments: minimum, maximum, mean, standard deviation, RMS, zero-crossing frequency, variance, median, kurtosis, skewness, and Shannon entropy. The class label is binary (Healthy vs. Seizure) and is perfectly balanced (100 samples each, 1:1 ratio). Despite its small size, DS3 provides a clean, separable feature space that isolates the effect of regularisation and preprocessing on a low-dimensional, balanced problem.

TABLE I: Dataset Comparison

Dataset	Samples	Features	Imbalance	Feature Type	Source
DS1 — UCI Epileptic	11,500	178	1:4	Raw EEG time-series	UCI ML Repository
DS2 — Clinical EEG	4,500	33	2.4:1	Clinical / demographic	Kaggle (EEG-FREE)
DS3 — Statistical EEG	200	11	1:1 (balanced)	Statistical features	Kaggle (newfeature)

III. METHODOLOGY

A. Preprocessing Pipelines

Two distinct preprocessing pipelines were designed and evaluated on all three datasets to test whether the ordering of preprocessing steps affects model performance.

Pipeline A: StandardScaler → SelectKBest (Mutual Information) → Logistic Regression.

Pipeline A first scales all features to zero-mean and unit-variance using StandardScaler, then selects the top-k most informative features using mutual information scoring (k = 30 for DS1, k = 20 for DS2, k = 8 for DS3). Scaling before feature selection ensures that the mutual information rankings are not distorted by feature magnitude.

Pipeline B: MinMaxScaler → PCA (95% variance) → Logistic Regression.

Pipeline B first scales all features to the [0, 1] range using MinMaxScaler, then reduces dimensionality via Principal Component Analysis (PCA) retaining components that explain 95% of total variance. This pipeline tests whether unsupervised dimensionality reduction can substitute for supervised feature selection, and whether the choice of scaler (MinMax vs. Standard) affects downstream performance.

B. Baseline Model

Logistic Regression with C = 1.0 (L2 regularisation, default) trained with up to 3,000 iterations (solver: lbfgs) was used as the baseline. The decision boundary is defined as:

$$P(y = 1 \mid x) = 1 / (1 + exp(-(\beta 0 + \beta^T x)))$$

Evaluation metrics include Accuracy, F1-Score, Precision, Recall, and Precision-Recall AUC (PR-AUC). PR-AUC is the primary metric given the class imbalance in DS1 and DS2, as accuracy is misleading in such settings.

C. Regularisation Study

Three regularisation strategies were compared: L1 (Lasso) with the SAGA solver, L2 (Ridge) with LBFGS, and Elastic Net (l1_ratio = 0.5) with SAGA. The regularisation objective is:

$$J(W, b) = (1/m)\ \Sigma\ L(\hat{y}, y) + (\lambda/2m)\ \Sigma||W||p$$

where p = 1 for L1, p = 2 for L2, and Elastic Net combines both. The parameter C (= 1/λ) controls regularisation strength. Feature sparsity was measured as the percentage of learned weights with absolute value below 10^{-4}.

D. Overfitting and Underfitting Demonstration

To demonstrate overfitting and underfitting behaviour, C was varied across [10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10, 100, 1000, 10000] on DS1 with Pipeline A preprocessing. Train F1 and Validation F1 were recorded at each value. Learning curves (F1 vs. training set size with 5-fold cross-validation) were also generated at C = 1.0.

E. Class Imbalance Handling

Four imbalance strategies were compared on DS2 (most clinically relevant imbalance): (1) No Handling (train as-is), (2) SMOTE oversampling of the minority class, (3) Random Undersampling of the majority class, and (4) Class Weighting (class_weight = 'balanced'). Medical priority was placed on maximising Recall, since a missed seizure is more clinically dangerous than a false alarm.

F. Comprehensive Comparison Grid

A full grid of 27 experiments was executed: 3 datasets × 3 regularisers (L1, L2, Elastic Net) × 3 imbalance strategies (None, SMOTE, Class Weighting). All configurations used Pipeline A preprocessing. Results were analysed by mean F1 across datasets and by per-dataset breakdown.

IV. RESULTS

A. Pipeline Comparison

Table II summarises the results of Pipeline A vs. Pipeline B across all three datasets. The most significant difference was observed on DS2 (Clinical EEG Features): Pipeline A achieved perfect scores (Accuracy = 1.000, F1 = 1.000, PR-AUC = 1.000) while Pipeline B produced noticeably degraded performance (Accuracy = 0.704, F1 = 0.827, PR-AUC = 0.703). On DS1 and DS3, performance was comparable between the two pipelines.

TABLE II: Pipeline A vs. Pipeline B Results

Dataset	Pipeline	Acc	F1	Prec	Recall	PR-AUC
DS1 — UCI (178 feat.)	Pipeline A	0.805	0.051	1.000	0.026	0.445
DS1 — UCI (178 feat.)	Pipeline B	0.804	0.043	1.000	0.022	0.453
DS2 — Clinical (33 feat.)	Pipeline A	1.000	1.000	1.000	1.000	1.000

Dataset	Pipeline	Acc	F1	Prec	Recall	PR-AUC
DS2 — Clinical (33 feat.)	Pipeline B	0.704	0.827	1.000	1.000	0.703
DS3 — Statistical (11 feat.)	Pipeline A	1.000	1.000	1.000	1.000	1.000
DS3 — Statistical (11 feat.)	Pipeline B	0.975	0.974	1.000	0.950	1.000

B. Overfitting and Underfitting

Table III shows the Train F1 and Validation F1 across the full range of C values on DS1. At $C = 10^{-5}$ and $C = 10^{-4}$, both train and validation F1 are 0.0000, confirming severe underfitting: the model is too constrained to learn any discriminative pattern. From $C = 10^{-3}$ to $C = 1$, the model begins to learn, but F1 remains low on both sets, indicating that the feature space of DS1 is inherently challenging for a linear classifier regardless of regularisation. No classic overfitting gap (high train, low val) was observed, which suggests that the dataset's EEG time-series features lack strong linear separability rather than that the model is memorising noise.

TABLE III: Overfitting / Underfitting — Train vs Validation F1 (DS1)

C Value	Train F1	Val F1	Status
1e-05	0.0000	0.0000	UNDERFIT
1e-04	0.0000	0.0000	UNDERFIT
0.001	0.0087	0.0000	UNDERFIT
0.01	0.0519	0.0258	UNDERFIT
0.1	0.0863	0.0426	UNDERFIT
1	0.1001	0.0508	Low / Borderline
10	0.1040	0.0550	Borderline
100	0.1050	0.0550	Borderline
1000	0.1060	0.0550	Borderline
10000	0.1050	0.0550	Borderline

C. Regularisation Comparison

Table IV presents the regularisation results across all three datasets. The most important pattern concerns feature sparsity: on DS2, L1 drove 95.0% of the 20 selected features to zero, compared to 35.0% for Elastic Net and 0.0% for L2. On DS3, L1 produced 75.0% sparsity vs. 37.5% for Elastic Net. Despite this extreme sparsity, L1 still achieved perfect F1 on DS2 and DS3, confirming that only a small number of features carry nearly all the discriminative signal in these datasets.

On DS1 (most challenging, 178 EEG features), all three regularisers produced similarly low F1 scores (0.034–0.038), confirming that the challenge is not regularisation choice but rather the fundamental difficulty of linearly separating raw EEG time-series features. PR-AUC values of ~0.441 indicate that the model's probability outputs are meaningfully calibrated even where hard classification fails.

TABLE IV: Regularisation Results Across Datasets (Pipeline A, C=1.0)

Dataset	Regulariser	Acc	F1	Prec	Recall	PR-AUC	Sparse%

Dataset	Regulariser	Acc	F1	Prec	Recall	PR-AUC	Sparse%
DS1 (UCI)	L1 (Lasso)	0.803	0.034	1.000	0.017	0.441	3.3%
DS1 (UCI)	L2 (Ridge)	0.804	0.038	1.000	0.020	0.441	0.0%
DS1 (UCI)	Elastic Net	0.804	0.038	1.000	0.020	0.441	0.0%
DS2 (Clinical)	L1 (Lasso)	1.000	1.000	1.000	1.000	1.000	95.0%
DS2 (Clinical)	L2 (Ridge)	1.000	1.000	1.000	1.000	1.000	0.0%
DS2 (Clinical)	Elastic Net	1.000	1.000	1.000	1.000	1.000	35.0%
DS3 (Statistical)	L1 (Lasso)	0.975	0.976	0.952	1.000	1.000	75.0%
DS3 (Statistical)	L2 (Ridge)	1.000	1.000	1.000	1.000	1.000	0.0%
DS3 (Statistical)	Elastic Net	1.000	1.000	1.000	1.000	1.000	37.5%

D. Class Imbalance Handling

On DS2, all four imbalance strategies achieved perfect metrics (Precision = Recall = F1 = PR-AUC = 1.000). This outcome, while unexpected, reflects the highly structured and linearly separable nature of DS2's clinical features: the 20 SelectKBest features carry almost complete discriminative information about epilepsy type, making the class imbalance correction largely irrelevant for this particular dataset. This finding itself is analytically important: when features are highly informative and the signal is strong, imbalance handling methods converge to equivalent performance, and the principal challenge lies in feature quality rather than class distribution.

The medical priority of maximising Recall (minimising missed seizures) is noted as a design principle for threshold selection in production systems, even where aggregate metrics appear equivalent.

E. Comprehensive Grid Results

Table V summarises the 27-experiment grid averaged across datasets. Mean F1 scores are consistently high on DS2 and DS3 regardless of configuration, while DS1 remains the hardest dataset. The interaction between regularisation and imbalance handling is most visible on DS1, where SMOTE and Class Weighting both substantially improve Recall (from ~0.02 to ~0.44) at a cost of Accuracy (from ~0.80 to ~0.65), confirming the fundamental Precision-Recall trade-off.

TABLE V: Full Comparative Grid — Selected Results (DS1, Hardest Dataset)

Dataset	Regulariser	Imbalance	Acc	F1	Recall	PR-AUC
DS1	L1	None	0.8048	0.0467	0.0239	0.4334
DS1	L1	SMOTE	0.6439	0.3292	0.4370	0.4226
DS1	L1	Weighted	0.6678	0.3414	0.4304	0.4216
DS1	L2	None	0.8052	0.0508	0.0261	0.4324
DS1	L2	SMOTE	0.6430	0.3298	0.4391	0.4221
DS1	L2	Weighted	0.6661	0.3391	0.4283	0.4219
DS1	EN	None	0.8048	0.0467	0.0239	0.4332
DS1	EN	SMOTE	0.6435	0.3301	0.4391	0.4221
DS1	EN	Weighted	0.6648	0.3382	0.4283	0.4218

V. DISCUSSION

A. RQ1: Does Preprocessing Order Affect Results?

Yes, significantly — particularly on DS2. Pipeline A's use of supervised feature selection (SelectKBest with mutual information) preserves the most clinically discriminative features before passing them to the classifier. Pipeline B's PCA projection creates orthogonal components that maximise variance rather than class separability, which can discard clinically relevant but low-variance features. On DS2, this resulted in a 29.6% drop in accuracy and a 17.3% drop in PR-AUC for Pipeline B. The choice of scaler (StandardScaler vs. MinMaxScaler) also affects PCA, as MinMax scaling compresses all features to [0, 1], which alters the covariance structure that PCA decomposes. On DS1 and DS3, both pipelines performed similarly, suggesting that preprocessing order matters most when features have heterogeneous importance levels and when class imbalance interacts with the signal distribution.

B. RQ2: Which Regularisation Generalises Best?

L2 (Ridge) generalises most consistently across all three datasets. It achieved equal or higher F1 than L1 on DS1 (0.038 vs. 0.034) and DS3 (1.000 vs. 0.976), and matched L1 perfectly on DS2. L2 does not produce sparse models, which is appropriate when features selected by SelectKBest are all considered informative. L1's tendency to zero out coefficients aggressively was beneficial on DS2 (demonstrating that only a few clinical features suffice) but did not improve final predictive performance compared to L2. L2 is recommended as the default regulariser for EEG classification with Logistic Regression.

C. RQ3: Does Elastic Net Consistently Outperform L1 and L2?

No. Elastic Net did not consistently outperform either L1 or L2 across the three datasets. In all configurations on DS2 and DS3, Elastic Net matched L2 exactly. On DS1, Elastic Net matched L2 and both slightly outperformed L1. The moderate sparsity produced by Elastic Net (35.0% on DS2, 37.5% on DS3) sits between the extremes of L2 (0%) and L1 (95%, 75%), but this intermediate sparsity offered no performance advantage. Elastic Net is most valuable when both feature selection and stable generalisation are needed simultaneously, but requires careful tuning of the l1_ratio hyperparameter, adding complexity without consistent gain in this study.

D. RQ4: How Does Imbalance Handling Interact With Regularisation?

On DS1 (the only dataset where imbalance handling meaningfully changed results), SMOTE and Class Weighting both improved Recall substantially (from ~2–3% to ~43–44%) at a consistent cost to Accuracy (~80% to ~64–68%) and Precision (1.00 to ~0.26–0.28). This trade-off was uniform across all three regularisers (L1, L2, Elastic Net), indicating that on this dataset, imbalance strategy dominates regularisation choice in determining the Precision-Recall balance. SMOTE and Class Weighting produced almost identical PR-AUC values (~0.42), suggesting that synthesising minority samples and re-weighting loss are equivalent strategies when the feature space is linearly non-separable. For medical applications prioritising seizure detection Recall, Class Weighting is preferred as it avoids introducing synthetic samples that may not reflect real seizure physiology.

E. Limitations

Several limitations should be noted. First, DS2 and DS3 produced near-perfect results across all configurations, which limits the interpretability of comparative analyses on those datasets and may indicate data leakage or overly separable feature spaces. Second, the study is limited to Logistic Regression; non-linear models such as Random Forest or SVM may reveal different sensitivity to preprocessing choices. Third, DS3 with only 200 samples is too small for robust generalisation claims. Fourth, the absence of temporal cross-validation (e.g., patient-specific splits) in DS1 may inflate performance estimates relative to real clinical deployment.

VI. CONCLUSION

This study conducted a systematic experimental analysis of how preprocessing choices, regularisation strategy, and class imbalance handling affect Logistic Regression generalisation on three real-world EEG seizure datasets. The key findings are summarised as follows:

- Preprocessing order matters: Pipeline A (supervised feature selection) significantly outperforms Pipeline B (PCA) on the clinical EEG dataset, with a 29.6% accuracy gap and 29.7% PR-AUC gap.
- L2 regularisation generalises most consistently across all datasets and is the recommended default for EEG-based seizure classification with Logistic Regression.
- Elastic Net does not consistently outperform L1 or L2 and adds hyperparameter complexity. L1 produces informative sparsity patterns but underperforms L2 in accuracy on DS3.
- Imbalance handling (SMOTE or Class Weighting) dramatically improves Recall on DS1, which is critical for clinical seizure detection. The interaction with regularisation is secondary to imbalance strategy on high-imbalance datasets.
- PR-AUC is a more informative evaluation metric than Accuracy for imbalanced EEG classification tasks, as accuracy can be misleadingly high when the majority class dominates.

Future work should explore non-linear classifiers, patient-independent cross-validation protocols, and multi-channel EEG feature extraction to improve generalisation in real clinical environments.

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